

# # Chapter 9 - Control Operations, Exceptions, and Certification

## ## 1. Document control

Status: Owner-authorized working amendment; recertification pending. Control families: GOV-007 GOV-008 DOC-002 OPS-001.  
Version: 1.1. Source authority: Book I then Book II.

## ## 2. Purpose

This chapter defines enforceable common engineering standards for control catalog, checklists, templates, retention, tooling, automation, reporting, exception forms, engineering certification.

## ## 3. Scope

Applies to every HAL contributor, change, component, provider adapter, interface, environment, and release within this subject area.

## ## 4. Authority and source requirements

Book I Decisions 25, 40, 50, 58. Book II 25, 29, 30, 35. If this chapter conflicts with either source, the higher-order requirement governs; record the conflict and correct Book III.

## ## 5. Definitions

Conformance evidence is a durable, reviewable record. A consequential change affects authority, identity, protected state, trust, privacy, a Reality Boundary, availability, or a material outcome. A control exception is a time-bounded approval, never a silent permission.

## ## 6. Normative standards

Each consequential control MUST have a stable ID, owner, applicability, enforcement, evidence, severity, source mapping, and automation status. Exceptions MUST be time-bounded and include justification, scope, risk, compensating controls, approver, effective/expiry/review dates, evidence, and revocation conditions.

## ## 7. Required engineering practices

Teams MUST perform design, security, privacy, reliability, and verification work in proportion to risk. Teams MUST retain evidence in the change or release record and MUST update traceability when requirements, interfaces, controls, or evidence change.

## ## 8. Prohibited practices

Permanent silent exceptions, constitutional waivers, and expired exceptions that continue to permit live-effect behavior are prohibited.

## ## 9. Required evidence

Catalog record, exception record, control report, certification report, retention index.

## ## 10. Automated enforcement

Control-as-code where practical; exception-expiry detection; periodic certification sampling.

## ## 11. Human review requirements

Control owner maintains controls; exception authority is limited by the catalog and never exceeds Book I.

## ## 12. Exceptions and waiver authority

An exception MUST identify affected control, justification, scope, risk, compensating controls, approver, effective date, expiration date, review date, evidence, and revocation conditions. Constitutional invariants cannot be waived. Architecture deviations follow the architecture-governance process. Expiration MUST fail closed or trigger explicit escalation.

## **## 13. Failure consequences**

A violated MUST control is a finding. Critical or high findings block merge or release until corrected, contained, or covered by an active approved exception. A detected authority, privacy, or trust-boundary failure MUST be contained and investigated.

## **## 14. Security considerations**

Controls MUST preserve explicit identity, authority, provenance, and least privilege; security controls protect HAL, while authority controls also prevent HAL from exceeding its mandate.

## **## 15. Privacy considerations**

Evidence and telemetry MUST minimize personal and sensitive data, use classification-aware access, and retain only what is required for the stated purpose and governing obligation.

## **## 16. Reliability considerations**

Critical behavior MUST define hazard-bounded degradation, containment, recovery, and evidence preservation. Applicable safety invariants, authority controls, and audit standards MUST NOT be degraded under resource pressure.

## **## 17. Verification method**

Verify through automated checks and risk-scaled test/release evidence, plus independent review or the documented Solo-Owner Assurance Profile where Book III Chapter 8 §11.1 permits it. Critical invariants require an identified repeatable verification method; critical failure modes require tested containment or recovery.

## **## 18. Metrics**

Track control pass rate, finding severity and age, exception count and age, verification coverage by risk, rollback/recovery performance, and post-release conformance defects.

## **## 19. Traceability to Book I**

Book I Decisions 25, 40, 50, 58.

## **## 20. Traceability to Book II**

Book II 25, 29, 30, 35.

## **## 21. Examples**

Example: the change record links the control IDs, source locators, test result, and approval before a protected action is released.

## **## 22. Anti-patterns**

Anti-pattern: a technically successful change is released without authority-path verification because it appears operationally routine.

## **## 23. Review findings**

Initial constitutional, architecture, enforceability, testability, security, privacy, reliability, usability, automation, exception-safety, duplication, and contradiction review: no unresolved internally resolvable issue. Reassess when implementation evidence is available.

## **## 24. Owner Review items**

None. Routine implementation choices are resolved through engineering judgment. Escalate only matters reserved by Book I.

## **## 25. Completion status**

Draft complete; chapter review record required before certification.